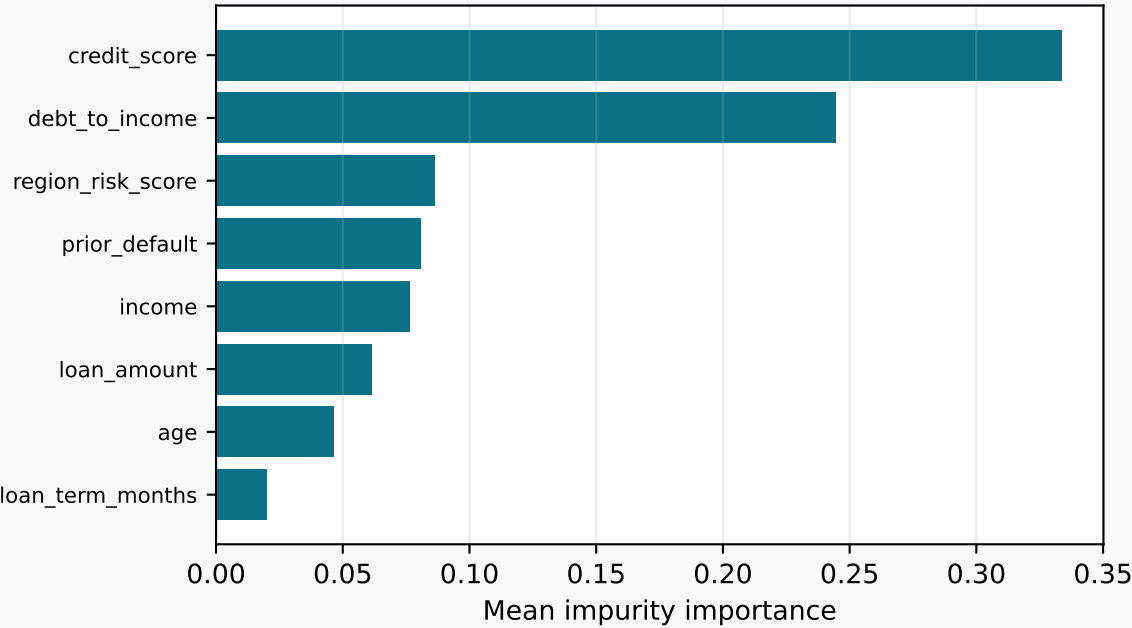


Task 3 - Model Benchmark Comparison

Primary model choice is a governance decision, not only an accuracy contest.

Model	AUC	Accuracy	Test N	Governance role
Logistic regression	0.800	0.876	1500	Primary model for adverse-action mapping and auditability.
Random Forest	0.793	0.785	1500	Challenger benchmark for non-linear signal and explanation contrast.

Random Forest challenger: top feature importances



Interpretation:

- Logistic regression is kept as the primary model because it gives stable, signed feature contributions that can be mapped to consumer-facing adverse-action reasons.
- Random Forest is included as a challenger to test non-linear predictive structure. It can improve modelling flexibility, but its native feature importances are less suitable for individual consumer notices.
- A production version could add TreeSHAP for the challenger model, but the submitted governance workflow deliberately prioritizes explainability, auditability and reason quality.

Conclusion: the non-linear benchmark strengthens technical diligence, while the chosen primary model remains aligned with fair-lending explanation duties.